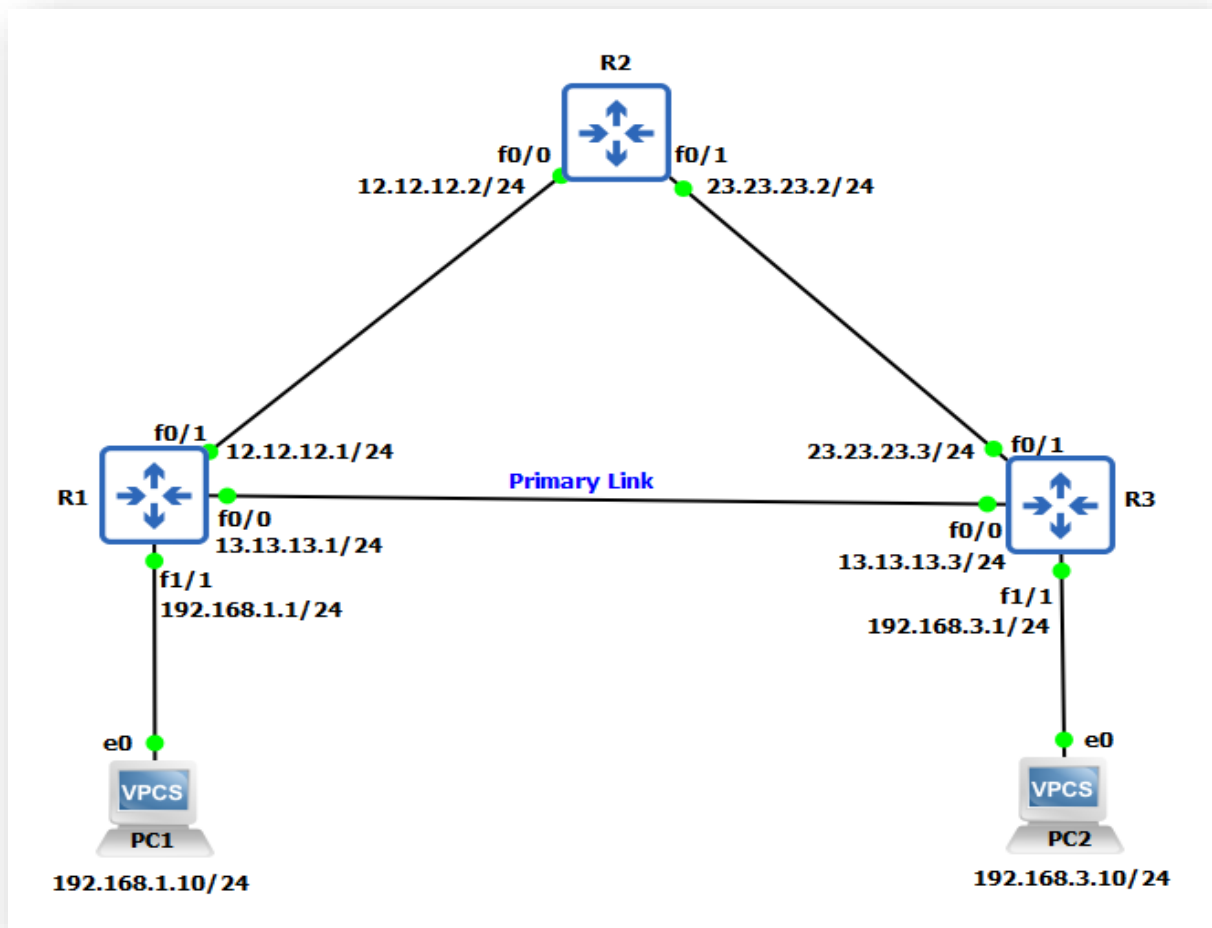


Static Floating Routing



Task

1. Configure routers with IP address as shown in topology and configure enable password as **ccna** and device access password as **cisco** (Refer Lab-1)
2. Configure static routes (next-hop) on R1, R2 and R3.
3. Configure floating route on R1 for R3's LAN network with Administrative Distance of 10
4. Configure floating route on R3 for R1's LAN network with Administrative Distance of 10
5. PC1 & PC2 should ping each other.

Static routing on R1 Router

```
R1#config t
R1(config)#ip route 192.168.3.0 255.255.255.0 13.13.13.3
R1(config)#ip route 23.23.23.0 255.255.255.0 12.12.12.2
R1(config)#exit
R1#
```

Static routing on R2 Router

```
R2#config t
R2(config)#ip route 192.168.1.0 255.255.255.0 12.12.12.1
R2(config)#ip route 192.168.3.0 255.255.255.0 23.23.23.3
R2(config)#exit
R2#
```

Static routing on R3 Router

```
R3#config t
R3(config)#ip route 192.168.1.0 255.255.255.0 13.13.13.1
R3(config)#ip route 12.12.12.0 255.255.255.0 23.23.23.2
R3(config)#exit
R3#
```

✓ Verification & Testing:

R1#sh ip route static

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override
 Gateway of last resort is not set

23.0.0.0/24 is subnetted, 1 subnets

```
S    23.23.23.0 [1/0] via 12.12.12.2
S    192.168.3.0/24 [1/0] via 13.13.13.3
R1#
```

R3#sh ip route static

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 + - replicated route, % - next hop override

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets

```
S    12.12.12.0 [1/0] via 23.23.23.2
S    192.168.1.0/24 [1/0] via 13.13.13.1
R3#
```

R1#ping 192.168.3.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/60/84 ms

R1#

R3#ping 192.168.1.10

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 44/59/76 ms

R3#

PC1> ping 192.168.3.10

192.168.3.10 icmp_seq=1 timeout

192.168.3.10 icmp_seq=2 timeout

84 bytes from 192.168.3.10 icmp_seq=3 ttl=62 time=38.409 ms

84 bytes from 192.168.3.10 icmp_seq=4 ttl=62 time=35.368 ms

84 bytes from 192.168.3.10 icmp_seq=5 ttl=62 time=40.513 ms

PC1> trace 192.168.3.10

trace to 192.168.3.10, 8 hops max, press Ctrl+C to stop

1 192.168.1.1 9.076 ms 9.383 ms 9.166 ms

2 13.13.13.3 31.124 ms 31.154 ms 31.264 ms

3 *192.168.3.10 41.118 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=62 time=41.309 ms

84 bytes from 192.168.1.10 icmp_seq=2 ttl=62 time=36.486 ms

84 bytes from 192.168.1.10 icmp_seq=3 ttl=62 time=33.602 ms

84 bytes from 192.168.1.10 icmp_seq=4 ttl=62 time=38.390 ms

84 bytes from 192.168.1.10 icmp_seq=5 ttl=62 time=44.502 ms

PC2> trace 192.168.1.10

trace to 192.168.1.10, 8 hops max, press Ctrl+C to stop

1 192.168.3.1 7.525 ms 10.276 ms 9.244 ms

2 13.13.13.1 30.309 ms 31.337 ms 32.274 ms

3 *192.168.1.10 41.254 ms (ICMP type:3, code:3, Destination port unreachable)

Configure floating route on R1 for R3's LAN network & on R3 for R1's LAN network with AD (Administrative Distance) of 10.

Floating route on R1 Router

R1#config t

R1(config)#ip route 192.168.3.0 255.255.255.0 12.12.12.2 10

R1(config)#exit

R1#

Floating route on R3 Router

```
R3#config t
R3(config)#ip route 192.168.1.0 255.255.255.0 23.23.23.2 10
R3(config)#exit
R3#
```

✓ Verification & Testing:**R1#sh run | section ip route**

```
ip route 23.23.23.0 255.255.255.0 12.12.12.2
ip route 192.168.3.0 255.255.255.0 13.13.13.3
ip route 192.168.3.0 255.255.255.0 12.12.12.2 10
R1#
```

R1#sh ip route static | begin Gateway

Gateway of last resort is not set

```
23.0.0.0/24 is subnetted, 1 subnets
S    23.23.23.0 [1/0] via 12.12.12.2
S    192.168.3.0/24 [1/0] via 13.13.13.3
R1#
```

R1#sh ip route 192.168.3.10

```
Routing entry for 192.168.3.0/24
  Known via "static", distance 1, metric 0
  Routing Descriptor Blocks:
    * 13.13.13.3
      Route metric is 0, traffic share count is 1
R1#
```

R1#traceroute 192.168.3.10

```
Type escape sequence to abort.
Tracing the route to 192.168.3.10
VRF info: (vrf in name/id, vrf out name/id)
  1 13.13.13.3 60 msec 64 msec 60 msec
  2 192.168.3.10 2552 msec 56 msec 68 msec
R1#
```

R3#sh run | section ip route

```
ip route 12.12.12.0 255.255.255.0 23.23.23.2
ip route 192.168.1.0 255.255.255.0 13.13.13.1
ip route 192.168.1.0 255.255.255.0 23.23.23.2 10
R1#
```

R3#sh ip route static | begin Gateway

Gateway of last resort is not set

```
12.0.0.0/24 is subnetted, 1 subnets
S    12.12.12.0 [1/0] via 23.23.23.2
S    192.168.1.0/24 [1/0] via 13.13.13.1
```

R1#

R3#sh ip route 192.168.1.10

Routing entry for 192.168.1.0/24

Known via "static", distance 1, metric 0

Routing Descriptor Blocks:

* 13.13.13.1

Route metric is 0, traffic share count is 1

R3#

R3#traceroute 192.168.1.10

Type escape sequence to abort.

Tracing the route to 192.168.1.10

VRF info: (vrf in name/id, vrf out name/id)

1 13.13.13.1 60 msec 56 msec 56 msec

2 192.168.1.10 2580 msec 56 msec 52 msec

R3#

Simulate the primary link failure by shutting down interface fa0/0 on R1 & R3 router.

R1 router configuration:

R1#config t

R1(config)#int fa0/0

R1(config-if)#shut

R1(config-if)#exit

R1(config)#exit

R1#

***Oct 10 17:19:54.455: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down**

R3 router configuration:

R3#config t

R3(config)#int fa0/0

R3(config-if)#shut

R3(config-if)#exit

R3(config)#exit

R3#

***Oct 10 17:19:57.911: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down**

✓ **Verification & Testing:**

R1#sh ip int brief | exclude unassigned

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	13.13.13.1	YES	manual	administratively down	down
FastEthernet0/1	12.12.12.1	YES	manual	up	up
FastEthernet1/1	192.168.1.1	YES	manual	up	up

R1#

R3#sh ip int brief | exclude unassigned

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	13.13.13.3	YES	manual	administratively down	down
FastEthernet0/1	23.23.23.3	YES	manual	up	up
FastEthernet1/1	192.168.3.1	YES	manual	up	up

R3#

R1#sh ip route static | begin Gateway

Gateway of last resort is not set

23.0.0.0/24 is subnetted, 1 subnets

S 23.23.23.0 [1/0] via 12.12.12.2

S 192.168.3.0/24 [10/0] via 12.12.12.2

R1#

R1#sh ip route 192.168.3.10

Routing entry for 192.168.3.0/24

Known via "static", distance 10, metric 0

Routing Descriptor Blocks:

* 12.12.12.2

Route metric is 0, traffic share count is 1

R1#

R1#traceroute 192.168.3.10

Type escape sequence to abort.

Tracing the route to 192.168.3.10

VRF info: (vrf in name/id, vrf out name/id)

1 12.12.12.2 76 msec 76 msec 76 msec

2 23.23.23.3 120 msec 100 msec 100 msec

3 *

192.168.3.10 324 msec 80 msec

R1#

R3#sh ip route static | begin Gateway

Gateway of last resort is not set

12.0.0.0/24 is subnetted, 1 subnets

S 12.12.12.0 [1/0] via 23.23.23.2

S 192.168.1.0/24 [10/0] via 23.23.23.2

R3#sh ip route 192.168.1.10

Routing entry for 192.168.1.0/24

Known via "static", distance 10, metric 0

Routing Descriptor Blocks:

* 23.23.23.2

Route metric is 0, traffic share count is 1

R3#traceroute 192.168.1.10

Type escape sequence to abort.

Tracing the route to 192.168.1.10

VRF info: (vrf in name/id, vrf out name/id)

- 1 23.23.23.2 64 msec 60 msec 52 msec
- 2 12.12.12.1 80 msec 80 msec 80 msec
- 3 192.168.1.10 2548 msec 80 msec 80 msec

R3#

PC1> ping 192.168.3.10

84 bytes from 192.168.3.10 icmp_seq=1 ttl=61 time=57.088 ms
84 bytes from 192.168.3.10 icmp_seq=2 ttl=61 time=61.357 ms
84 bytes from 192.168.3.10 icmp_seq=3 ttl=61 time=65.261 ms
84 bytes from 192.168.3.10 icmp_seq=4 ttl=61 time=57.210 ms
84 bytes from 192.168.3.10 icmp_seq=5 ttl=61 time=56.359 ms

PC1> trace 192.168.3.10

trace to 192.168.3.10, 8 hops max, press Ctrl+C to stop

- 1 192.168.1.1 2.294 ms 10.287 ms 10.414 ms
- 2 12.12.12.2 32.266 ms 30.255 ms 31.242 ms
- 3 23.23.23.3 53.178 ms 53.135 ms 53.129 ms
- 4 *192.168.3.10 63.005 ms (ICMP type:3, code:3, Destination port unreachable)

PC2> ping 192.168.1.10

84 bytes from 192.168.1.10 icmp_seq=1 ttl=61 time=57.117 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=61 time=59.249 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=61 time=51.308 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=61 time=65.297 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=61 time=56.274 ms

PC2> trace 192.168.1.10

trace to 192.168.1.10, 8 hops max, press Ctrl+C to stop

- 1 192.168.3.1 9.285 ms 9.214 ms 8.360 ms
- 2 23.23.23.2 32.106 ms 31.089 ms 31.208 ms
- 3 12.12.12.1 53.119 ms 51.241 ms 53.360 ms
- 4 *192.168.1.10 64.653 ms (ICMP type:3, code:3, Destination port unreachable)

Important Commands:

```
sh ip route  
sh ip route static  
sh ip route static | begin Gateway  
sh ip int brief | exclude unassigned
```

Ratan